

TECHNICAL BRIEFING · LIVE DEMONSTRATION

The Company Brain

Orbital as a universal semantic data router

An engineering write-up of a working demonstration: **how it works, what was built** (fourteen live data sources behind one query), a **step-by-step trace**, a **live view of every tool**, the join into a **21.4-million-record UK distress corpus**, a tailored read on **Big Ideas Group's own portfolio**, and an honest **gap analysis**.

Prepared for Marty — Orbital · Sebastian Gray — Big Ideas Group

Subject `orbital-company-brain` · Engine `Orbital` (`orbitalhq.com`) · working live demo · 2026

Every result, latency and screenshot in this document was captured live from the running Orbital stack during one build session — not mocked. Where something is designed-but-incomplete it is marked `[gap]` and explained in §7.

Executive summary

The concept is a **semantic data router**. Rather than a human pre-building a connection between every pair of systems, each source is described once — by the *meaning* of the data it exchanges — and the engine ([Orbital](#)) works out which sources to call, in what order, to answer a request. You ask for a result by what it *is*; the router finds the path, runs it, and emits the lineage of how it was built.

This briefing is deliberately honest about proven-vs-designed:

- **It works, live, across fourteen sources.** From one company name the brain assembles identity, background, geography, crime, flood, weather, **real insolvency & ownership-risk**, **adverse media** and an LLM-written narrative — composing services that were never told they connect. Measured warm latencies run **0.01–0.97 s** per capability (§4, Fig. 1).
- **The mechanism is a network effect, not plumbing.** Sources compose through shared semantic types. One ~30-line adapter (Postcodes.io) fused two clusters; adding the Nth source is $O(N)$, not $O(N^2)$ (Fig. 3).
- **The differentiated value is distress intelligence.** Because the brain derives a company's Companies House number (via GLEIF, no key), it joins straight into the author's **gcloud-intel** corpus — 21.4M records of Gazette insolvency + Companies House + PSC ownership + government contracts — to add real insolvency stage, **phoenix-director risk** and live **contracts-at-risk** (§5).
- **We ran it on Big Ideas Group's own world.** The brain mapped BIG's ownership chain, screened its portfolio and its founder for serial-failure risk (clean), and surfaced a textbook phoenix case — a company that *passes* a status check whose director controls 145 other distressed companies (§6).

One-line thesis. Every source you teach the brain makes every question smarter — at linear cost — and every answer comes with a receipt. A pile of point-to-point integrations can never become that.

A note on terms (plain English)

Term	What it actually means
Semantic type	A named <i>meaning</i> for a value — not <code>string</code> but <code>CompanyName</code> , <code>Postcode</code> . Two APIs that bind the same type can be joined automatically.
Taxi / Orbital	Taxi is the open language that describes each API in semantic types; Orbital is the engine that plans and runs the multi-API call graph. Taxi is open; the engine is the product.
TaxiQL	The query: <code>given { a } find { B }</code> — "I have an A, get me a B." You never name the APIs.
Adapter	One small Taxi file describing a single source: a response model with <code>jsonPath</code> bindings + an HTTP operation.
Lineage	The auditable graph Orbital emits for every answer — which fact entered, which services ran, in what order.
Identity spine	The shared key (<code>CompanyRegistrationNumber</code>) that lets one company be recognised across every source.
GLEIF / PSC	GLEIF: the free global registry — name → LEI + Companies House number, no key. PSC: "Persons with Significant Control" — who owns/controls each UK company.

1. How it works

Four ideas. **(1) Shared semantic types** — each API's fields are bound to *meanings*; Police, TfL and Open-Meteo all bind the same `Latitude / Longitude` without knowing about each other. **(2) Describe, don't integrate** — an adapter is a few lines; there is no point-to-point code between any two APIs. **(3) Query-time composition** — Orbital matches the output type of one operation to the input of another and builds the plan; from a name alone it *derives* name → GLEIF → postcode → coordinates → crime. **(4) Lineage as proof** — every answer carries an auditable graph of how it was built.

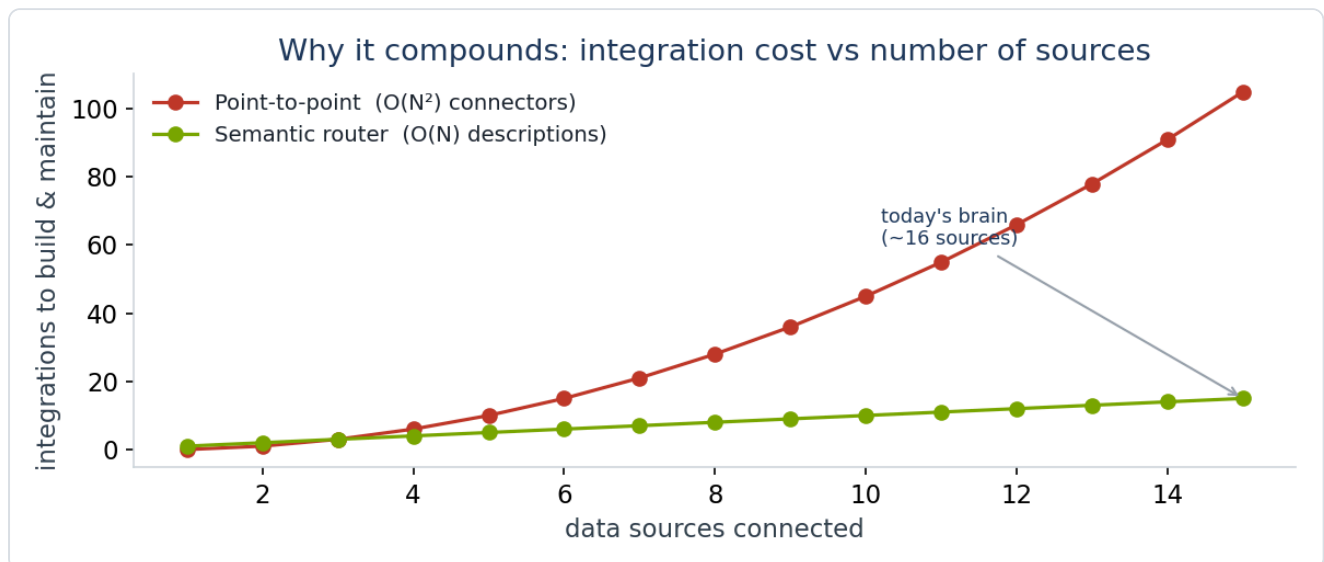


Figure 3. Point-to-point integration cost grows as $O(N^2)$; semantic descriptions grow as $O(N)$. At ~16 sources the brain would already be ~120 hand-built connectors under the old model.

Why it compounds. The Nth source becomes reachable from queries involving all N-1 prior ones, and enriches them, at $O(N)$ description cost. That inversion is the moat.

2. What we built

On a base of 9 callable UK-gov APIs, this session added the keystone bridge and a set of enrichment, distress, markets, code and personal adapters — **fourteen live sources** behind one query. Each adapter is a single self-contained Taxi file; none required editing another.

Adapter	From → to (semantic)	Auth	Role
Postcodes.io ★	Postcode → Lat/Lng	none	Keystone — fuses identity ↔ geo
GLEIF	CompanyName → LEI , CH number , Postcode	none	Name → identity; the spine key
Insolvency (gcloud-intel)	CH number → distress profile	none*	Distress · phoenix risk · contracts
Adverse media / social	CompanyName → news+social	none	Google News · HN · Reddit · Bluesky
LLM narrative	CompanyName → narrative	local	On-device LM Studio (qwen3)
Wikidata · Open-Meteo	name → description · Lat/Lng → weather	none	Background · context at HQ
Police.uk · EA Flood	Lat/Lng → crime · flood	none	Location risk signals
Yahoo · SEC EDGAR	Ticker → quote · CIK → filings	none	Markets vertical
GitHub repo + issues	owner/repo → facts + issues/PRs	none	Coding vertical
Personal meeting brief	calendar → attendees → companies	mock	Personal vertical (privacy-safe)

* The insolvency adapter reads a local read-only service over the corpus; no third-party auth. Personal uses a privacy-safe mock calendar (real Google MCP is a documented swap). On top sits "**The Company Brain**" web app and a hardened, cached, rate-limited public-sandbox build.

3. A query, step by step

We ask for a company 360 by meaning; the request names no APIs:

```
given { name : uk.gov.CompanyName = "TESCO PLC" }  
find { identity, background, weatherAtHq, crimesNearHq, floodStationsNearHq }
```

1 — identity. Only GLEIF turns a `CompanyName` into registry identity → it is called; Wikidata runs in parallel for a description. **2 — the invented hop.** Crime and weather need `Lat/Lng` ; nothing produced those, but GLEIF emitted a `Postcode` and Postcodes.io converts it — Orbital inserts that call itself. **3 — fan out.** Police, the Environment Agency and Open-Meteo are called in parallel on the shared coordinates. **4 — assemble.** Five responses, projected into the requested shape.

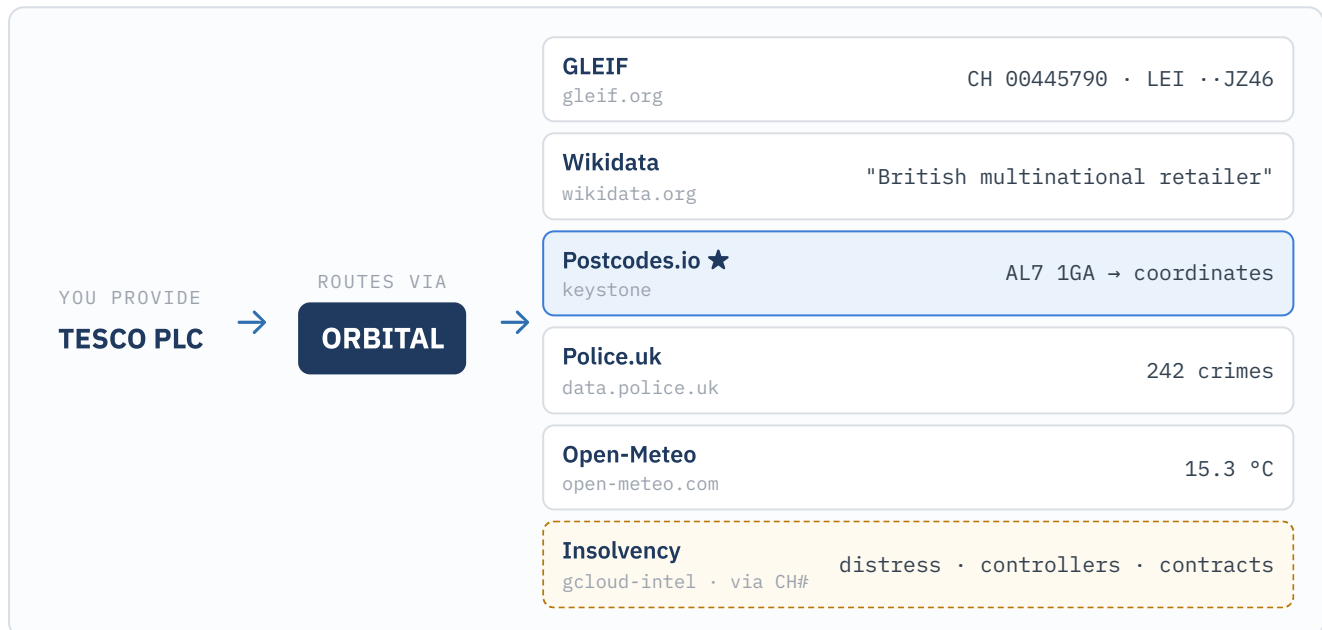


Figure 1b. The plan Orbital derived from one company name. The dashed node (§5) joins on the Companies House number GLEIF already produced — no new identity work.

Notice GLEIF produced `CH 00445790` . Any source keyed on a Companies House number is now reachable for this company at zero extra resolution cost — the hook for §5.

4. The tools — live views

A view of each capability, captured live this session. Latencies (warm) are in Figure 1.

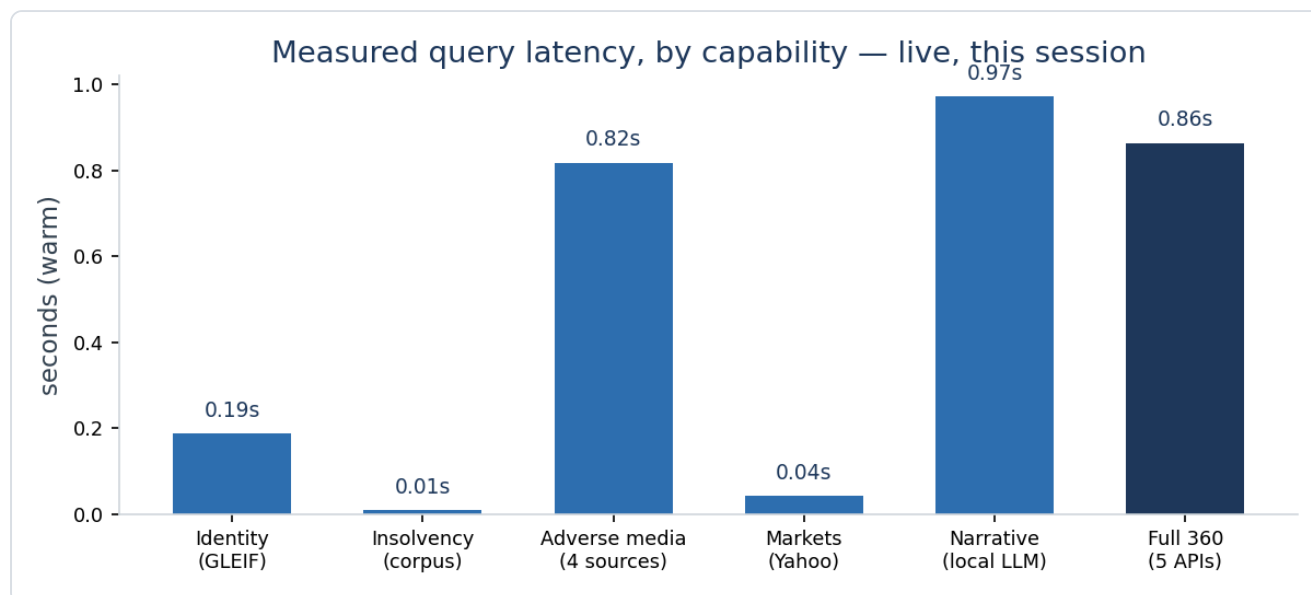


Figure 1. Measured warm latency per capability. Cached insolvency lookups return in ~10 ms; the local-LLM narrative is the slowest at ~1 s.

4.1 Company 360

```
given CompanyName="TESCO PLC" → find { ... }
```

5 APIs • 0.86 s

identity : TESCO PLC • CH 00445790 • LEI 2138002P5RNKC5W2JZ46 • ACTIVE • AL7 1GA
background : "British multinational retailer" (Wikidata Q487494)
weatherAtHq: 15.3 °C crimesNearHq: 242 floodStationsNearHq: 15

4.2 Insolvency & phoenix-director risk (the differentiated tool)

```
given CH#="07021926" → find { InsolvencyProfile }
```

gcloud-intel • 0.01 s

companyName : AASEYA SOFTWARE SERVICES (UK) LIMITED
stage : **Winding-up petition** latestEvent: winding-up-petition (2025-02-07)
controllerRisk : controller Manoj Kumar Baheti runs no other distressed companies on record
topContractBuyer : **Social Security Scotland**
topContractTitle : Adobe Creative Cloud, Document Cloud and Software

```
given CH#="10976115" → find { InsolvencyProfile }
```

the phoenix case

companyName : OFFICE TEK SOLUTIONS LTD
stage : No insolvency on record ← passes a status check
controllerRisk : **controller Mr Neville Taylor runs 145 other distressed companies**

The OFFICE TEK case is the point: the company itself is "Active" and clean — but its controller is a serial-failure phoenix. No Companies-House status check surfaces that; the PSC-network join does.

4.3 Adverse media / social

```
given CompanyName="TESCO PLC" → find { AdverseMediaList } news+social · 0.82 s
```

```
news      · "Preliminary Results 2025/26 – Tesco PLC"
news      · "Tesco faces slower first-quarter sales growth, says Citi" (Yahoo)
news      · "Tesco and Sainsbury gain comfort from Asda's continuing struggle"
hackernews · (matched stories)  reddit/bluesky · best-effort
counts: { news: 5, hackernews: 4, reddit: 0, bluesky: 0 }
```

4.4 LLM narrative (on-device)

```
given CompanyName="TESCO PLC" → find { RiskNarrative } local LM Studio · 0.97 s
```

"TESCO PLC, the UK's largest supermarket chain, faces mounting competitive and operational risks from intensifying discount retailers like Aldi and Lidl, rising input costs, and evolving consumer habits; its ongoing restructuring efforts ... aim to restore profitability but carry execution and reputational risks amid ongoing margin pressure." (qwen3-coder-next)

4.5 Markets · Coding · EDGAR · Personal

```
four more verticals, same router live
```

```
markets  AAPL → 306.31 USD (NMS)    · TSCO.L → 432.7 GBp (LSE)           [Yahoo]
coding    orbitalapi/orbital → 356★ · 13 issues · TypeScript
          open issue #31 "Add username tracking to query events" – @martypitt [GitHub]
filings   CIK 0000320193 → Apple Inc. · Electronic Computers · latest form 4, 2026-05-29 [SEC EDGAR]
personal  next meeting "Q3 partnership review – embedded payments" · Monzo / Revolut / Wise
```

The proof is the absence. None of these models declare an API name or an ordering. Every plan above is inferred from semantic types — which is exactly why a new source plugs in without touching anything that already works.

5. The gcloud-intel integration, explained

The demo so far enriches *public* identifiers. The step-change is to route the same query into a deep, proprietary corpus that already exists on disk: **gcloud-intel**, a working UK insolvency-and-procurement pipeline that fuses four sources into a local store and a local LLM.

5.1 The corpus

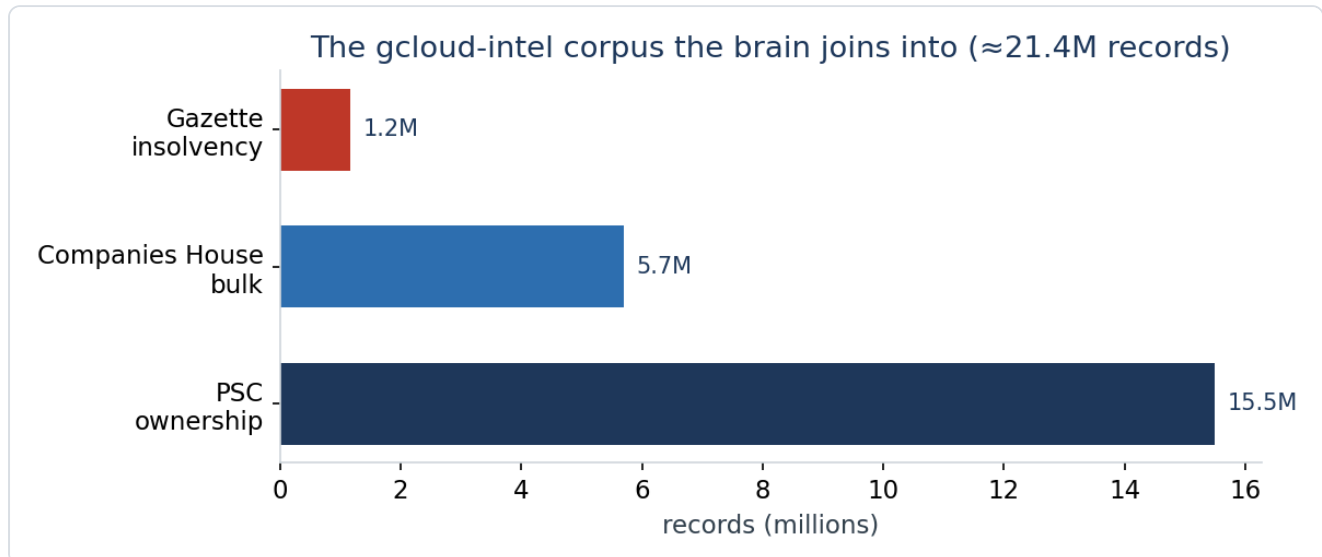


Figure 2. ≈21.4M records: Gazette insolvency notices, the full Companies House register, and 15.5M PSC ownership records — plus the G-Cloud/DOS contract corpus (2,496 suppliers mapped to CH numbers, 93% deterministically).

5.2 Why it joins instantly — the shared key

The reason this is an integration and not a project is the identity spine: the brain's GLEIF hop already produces a company's **Companies House number**, and gcloud-intel is keyed on the **Companies House number**. The join is deterministic — no fuzzy name-matching. Mechanically it is one adapter (a read-only service over the corpus + a ~15-line Taxi file) — built and verified live this session.

```
given { id : uk.gov.CompanyRegistrationNumber } // already produced by GLEIF
find { InsolvencyProfile } // stage · controller risk · contract at risk
```

5.3 What it adds (verified against the live corpus)

- **Insolvency stage.** CAVENDISH WOOD LTD (08810995) → liquidation / winding-up order; AASEYA (07021926) → winding-up petition.
- **Live government contracts at risk.** AASEYA holds a live **Social Security Scotland** contract — surfaced before internal teams notice; across the corpus, **~£30.6M of live gov-contract value** sits with suppliers in active insolvency, and those contracts re-compete within months (a sourcing signal for an investor).
- **Phoenix-director risk (PSC network).** **29,953 individuals** currently control 2+ distressed UK companies; the worst, Neville Taylor, sits on **146**. These names are in no other distress feed (§4.2, OFFICE TEK).

Data sovereignty, for free. gcloud-intel's inference runs on-device (LM Studio; no data leaves the host). With Orbital's deterministic, type-checked routing and native lineage, the result is a **local-first, fully-auditable KYB brain** — a strong posture for regulated buyers.

6. For Big Ideas Group

Big Ideas Group (CH# 04151059, active since 2001, EC1M 6DX) is a London multi-family office; founder **Sebastian Gray** is also a non-exec at **Napier** (financial-crime / AML screening). The brain is, for a backer, a **diligence and portfolio-risk engine** — and the distress/ownership layer is squarely in Sebastian's world. We pointed it at BIG's own world:

Run on BIG's world	What the brain returned (live)
BIG's ownership chain	BIG IDEAS GROUP LIMITED is 75–100% owned by Big Digital Ventures Limited — surfaced from PSC data.
Portfolio: Mutual Vision	Mutual Vision Technologies (04088165) — clean ; held via Mv Tech Bidco Limited + Conor Shiels (the PE deal structure, visible in ownership).
Phoenix screen — Sebastian Gray	Clean — absent from the 29,953-controller distress network. The reassuring answer a backer wants.
Phoenix capability proof	Worst controller Neville Taylor → 146 distressed companies ; a portfolio target's director would be screened against this before a cheque.

6.1 Four things BIG would use on Monday

- **Instant diligence** on any UK company — identity, ownership, distress, government contracts, adverse media — with lineage as the audit trail.
- **Phoenix-director screening** before backing or acquiring — the OFFICE TEK case (clean company, serial-failure director) is the pattern that off-the-shelf credit tools miss.
- **Distressed-supplier deal flow** — failing gov suppliers holding live contracts (£ at risk) that re-compete = a sourcing engine for acquisitions / re-compete plays.
- **Portfolio & watchlist monitoring** — re-run the dossier nightly; alert on distress, ownership or adverse-media changes across the portfolio and its suppliers.

Why this is also the best diligence on Orbital. Handing BIG a tool that answers their next deal question lets them *feel* the platform value — and watch a fourteenth source plug in without an integration project. Several BIG portfolio companies (Mutual Vision; the Napier-adjacent fintechs) could themselves build on the same engine — making Orbital infrastructure across the portfolio, not just one investment.

7. Honest gap analysis

What is weak, unproven, or would need work before production — in the spirit of a real technical review.

Gap	Status & what closes it
Coverage — GLEIF needs an LEI	The no-auth name→number path only covers LEI-registered firms. Closes with: a free Companies House search key for the millions of smaller UK companies (adapter already drafted).
Type coercion (found & fixed live)	Orbital failed to emit the insolvency model when fields used <code>inherits Boolean / Int</code> bound to JSON <code>true/2</code> . Fixed by using <code>String / Decimal</code> (the proven primitives). A real, documented sharp edge.
Insolvency perf & transport	Controller-risk recomputes over 15.5M PSC rows (~10–20 s cold) so we warm-cache demo companies (→10 ms). Production: materialise the PSC distress view + index. The service is reached via <code>host.docker.internal</code> ; native deploys swap to the host IP.
Source reliability	Police.uk is occasionally slow/empty; X/Twitter & LinkedIn are not free-scannable (paid social-listening needed); GDELT/Reddit are rate-limited. Closes with: caching, stubs, and a paid social provider where required.
It is a demo, not a product	No multi-tenant auth, RBAC, PII governance, or SLAs yet. Orbital has the policy primitives; they are not wired here. The local LLM narrative is on-device but unevaluated for accuracy.

The honest position. The architecture and the cross-source composition are real and measured. The data is real. What remains is hardening — coverage, performance, governance — not invention. That is a far better place to be than the reverse.

8. Where this goes next

- **Coverage:** Companies House search key (all UK firms) + the materialised PSC distress view for instant phoenix screening at any scale.
 - **Watch & alert:** re-run dossiers on a schedule; emit a signal when a counterparty's distress stage, ownership pattern, contract exposure or adverse-media changes.
 - **Ask-in-English:** gcloud-intel's local "Insolvency Oracle" (NL→SQL, on-device, audit-logged) is exactly the brain's LLM-as-a-source surface — wire it as a first-class query mode.
 - **Hosted sandbox:** the hardened, cached, rate-limited build ships as a "type any UK company" front door for Big Ideas Group.
 - **Governance:** turn on Orbital's per-type access policies + PII masking; lineage already gives the audit trail.
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Prepared from the live `orbital-company-brain` demo and the `gcloud-intel` corpus. Company, markets, coding, insolvency, adverse-media and narrative results are real responses captured during the build session; corpus and ownership figures are from the live `gcloud-intel` database. Orbital is the semantic data-integration engine behind the Taxi language (orbitalhq.com).